



A queueing model for time-dependent rental systems with phase-type distributed rentals and substitutions of items

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Abstract

Society currently experiences a growing interest in renting as an alternative to exclusive ownership (Fota et al., 2019; Yeganeh, 2021). This paper supports rental companies in this transition by providing a method to evaluate rental system occupancy. The method considers time-dependent systems in which customers rent substitutes during stockouts. Specifically, the paper provides a queueing model based on a continuous-time Markov chain for systems where rentals occur according to Poisson processes and rental times follow phase-type distributions. Additionally, the paper provides a heuristic optimization algorithm to minimize inventory capacity while maintaining an upper stockout probability limit. The paper evaluates these methods through numerical experiments, including a company case. The results indicate that the queueing model adequately reflects the occupancy distributions of the investigated items, and that the optimization algorithm minimizes the inventory capacity to near-optimality. In the company case, these methods reduce the inventory capacity while simultaneously improving the maximum stockout probability.

Keywords Queueing · Markov chain · Phase-type · Rental

1 Introduction

Renting is an established concept in the society and provides a popular alternative to exclusive ownership (Fota et al., 2019; Yeganeh, 2021). The benefits of renting include the ability to serve the same number of customers with fewer resources by pooling and sharing resources. With an increasing access to data and computational resources, rental companies can now optimize their inventories more effectively (Vassakis et al., 2018; Favaretto et al., 2023). This paper takes advantage of these opportunities by providing a method for turning data on arrival rates and rental times into occupancy estimates, focusing on cases where customers can rent substitutes during stockouts. Additionally, the paper provides an approach for optimizing the inventory capacity.

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The academic literature covers roughly 60 years of research on inventory replenishment and rental services (Wagner & Whitin, 1958; Tainiter, 1964; Jalali & Nieuwenhuyse, 2015; Ma et al., 2019). In more recent literature, Benjaafar and Hu (2020) investigate various sharing economies and derive three distinct types: The first is *Peer-to-peer Resource Sharing* (PRS), where the owner rents a single item to a non-owner, the second is *On-demand Service Platforms* (OSPs), where individual workers perform time-sensitive services, and the third is *On-demand Rental Networks* (ORNs), where a company provides on-demand rentals of items from several inventories to a large number of customers. Benjaafar and Hu (2020) emphasize that queueing theory is a convenient tool for describing the dynamics in rental systems and note that the appropriate modeling structure depends on the specific rental system. OSPs typically involve a random capacity since workers enter the platform voluntarily (Ibrahim, 2018; Dong & Ibrahim, 2020). Conversely, the ORNs involve a constant capacity for the whole system, whereas the capacity alternates in the individual queues. For instance, in car rental companies, a car might return to another vendor than the car was rented from. Finally, the PRS systems can be modeled using a single queue of constant capacity.

The academic literature contains a recurrent number of applications of queueing theory differing in model structure and scope. Tainiter (1964) conducted the first study describing a rental system using queueing theory and suggested to model the occupancy of items as a queue with customer loss and derive the profit of the system using the model. Researchers studied several similar cases in the following years. For instance, both Savin et al. (2005) and Gans and Savin (2007) investigated the problem of allocating capacity and optimizing profit for rental systems with two classes of customers. Specifically, Gans and Savin used a Markov decision process to decide on an action for each customer as they arrive at the queue. In addition, Papier and Thonemann (2008) described the occupancy of rail cars as a model with customer loss and derived the system's profit from the model. Inspired by the increasing interest in vehicle-sharing platforms, George and Xia (2011) derived a closed queueing network model for the occupancy of vehicles. Subsequently, the authors derived an approach for finding the optimal fleet size. In a slightly altered setting, Choi and Yoon (2016), Chakravarthy (2020) and Sivakumar et al. (2020) studied a problem they denoted the queueing-inventory system, where service-completions result in a loss of servers.

There seem to be abundant studies applying queueing theory to the rental industry, but many of them assume stationary customer behavior. In fact, this literature review uncovered only two studies accounting for time-dependent behavior. Papier and Thonemann (2008) evaluated the time-dependent behavior using the average stationary approximation method, and Epstein et al. (2020) used a Runge-Kutta method. Naturally, the wider literature on applied queueing theory contains several examples of time-dependent behavior, such as the analyses conducted by Izady and Worthington (2011) and Singh et al. (2021), and the investigations of hospital settings conducted by Bekker and Bruin (2010), Brummelen et al. (2018), and Andersen et al. (2019).

Returning to the rental industry, only Papier and Thonemann (2008) consider item substitution, and only three studies on item substitution exist in the wider literature on logistics and manufacturing (Ahiska et al., 2017; Walt & Bean, 2022; Mahmutoğullari & Yaman, 2023). A recent study by Mahmutoğullari and Yaman (2023) solved the problem of supplying retailers using mixed integer programming. In the study, retailers sell two product types, where one is of high quality, the other of low quality, and only the high-quality product can be used as a substitute.

This paper extends the literature on applied queueing theory by providing a time-dependent model where items can be substituted and customers follow phase-type (PH) distributed rental times. The paper limits the scope to rental systems consisting of a single inventory, where

customers arrive according to time-dependent Poisson processes and receive service on a first-come-first-served basis. Furthermore, the paper divides customers into types, where each type of customer prefers a type of items. Customers use the other items as substitutes, with a known probability, during stockouts of the preferred items.

In this paper, a Continuous-Time Markov Chain (CTMC) models the customers' behavior and evaluates the time-dependent state distribution using the uniformization method. A heuristic optimization algorithm supplements the CTMC by minimizing the total inventory capacity restricted by an upper stockout probability limit. The paper validates these approaches through numerical experiments and data from a rental company. In summary, the paper contributes to the literature on applied queueing theory with:

- A queueing model based on a CTMC for time-dependent rental systems. The model accommodates rental times following PH distributions and customers renting substitute items.
- A heuristic optimization algorithm for minimizing the total capacity of the inventory without violating an upper stockout probability limit.

In the following, Section 2 provides the details of the rental system. Next, Sections 3–4 define the above-mentioned Markov chain and optimization algorithm. Section 5 presents a numerical study, and Section 6 applies the methods to the company case. Finally, Section 7 presents the conclusions and suggestions for future work.

2 Problem description

This paper considers rental companies renting out items from an inventory, where the items return to the same inventory, such as household appliances, clothing, construction machines, and so on. Customers arrive and perform rental requests with random inter-arrival times. On arrival, a customer requests a single item, and if available, the customer rents and removes the item from the inventory for a random duration. When the rental period ends, the customer returns the item to the inventory. This paper considers rental companies serving customers on a first-come-first-served basis, as well as companies where the inventory capacity is sufficiently large that maintenance can be conducted ahead of the next rental. Furthermore, if an arriving customer requests an item during a stockout, the customer will attempt to rent a substitute with a known probability (cf. Figure 1). In this case, several substitutes might exist, and the preferences depend on the availability of all items in the inventory. If the customer decides *not* to rent a substitute, or the selected substitute is out of stock, the customer will leave the system.

Finally, demand fluctuations, such as seasonal behavior, causes the customer arrival rate and rental time to fluctuate over time. Thus, this paper views the system as a time-dependent queueing system with customer relocation, where the items correspond to the servers. Sections 2.1–2.2 elaborate on the formal details.

2.1 Arrivals and rental times

Let $\mathcal{C} = \{c_1, c_2, \dots, c_{|\mathcal{C}|}\}$ denote a set of customer types, and $\mathcal{A} = \{a_1, a_2, \dots, a_{|\mathcal{A}|}\}$ a set of item types, where each customer type exclusively prefers an item type of the same index. As a result, customer type c_1 prefers item type a_1 , customer type c_2 prefers item type a_2 , and so on. Further, let $|\mathcal{C}| = |\mathcal{A}| = n$ denote the number of customer and item types. This paper assumes that each customer type arrives at the inventory according to a time-dependent

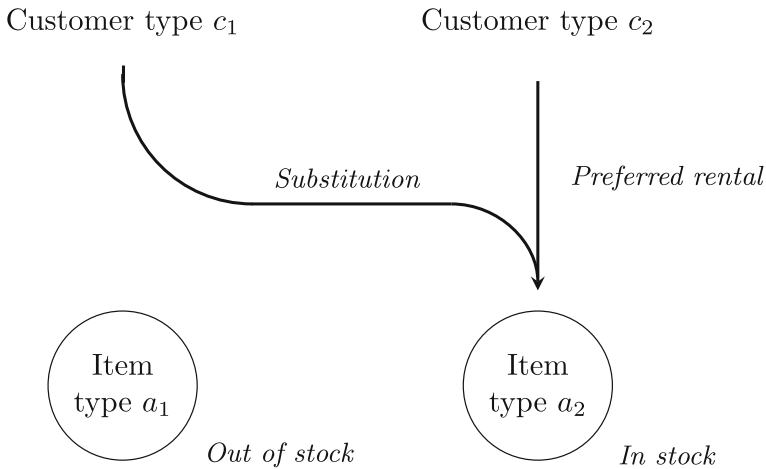


Fig. 1 An example of two customers and item types. A customer of type c_1 might choose to rent an item of type a_2 during a stockout

Poisson process with rate $\lambda_c(t) \in \mathbb{R}^+$, where $t \in \mathbb{R}$ denotes the time and $c \in \mathcal{C}$. Further, customers rent items with PH distributed rental time.

The PH distribution is a probability distribution defined by the time to absorption in a CTMC. The transition rate matrix of the absorbing CTMC has the form

$$\begin{pmatrix} \mathbf{\Gamma} & \boldsymbol{\gamma} \\ \mathbf{0} & 0 \end{pmatrix},$$

where the last row denotes the absorbing state. The matrix $\mathbf{\Gamma} \in \mathbb{R}^{o \times o}$ denotes the PH generator, the column vector $\boldsymbol{\gamma} = -\mathbf{\Gamma}\mathbf{1}$ defines the exit-rates, and $o \in \mathbb{N}$ the number of phases. Further, the row vector $(\boldsymbol{\beta}, 0)$, where $\boldsymbol{\beta} \in \mathbb{R}_{\leq 1}^o$, denotes a probability distribution, i.e. $\boldsymbol{\beta}\mathbf{1} = 1$, defining the initial distribution of the absorbing CTMC. Note that $\boldsymbol{\beta}$ is not to be confused with the beta distribution from the family of continuous probability distributions. Neuts (1981, p. 41) and Bladt and Nielsen (2017, p. 125) describe the PH distribution in details. This paper employs the PH distribution due to its ability to reflect a wide range of empirical distributions.

For the rental system in focus, let $(\boldsymbol{\beta}_{ca}(t), \mathbf{\Gamma}_{ca}(t))$ denote the representation of the PH distribution for customer type c renting item a at time t . In this representation, vector $\boldsymbol{\beta}_{ca}(t) \in \mathbb{R}_{\leq 1}^{|\mathcal{W}_{ca}|}$ denotes the initial distribution and matrix $\mathbf{\Gamma}_{ca}(t) \in \mathbb{R}^{|\mathcal{W}_{ca}| \times |\mathcal{W}_{ca}|}$ the PH generator, where $\mathcal{W}_{ca} \in \{1, 2, 3, \dots\}$ is a set of phases. Thus, $\boldsymbol{\gamma}_{ca}(t) = -\mathbf{\Gamma}_{ca}(t)\mathbf{1}$ denotes the exit-rate vector. The parameters $\beta_{caw}(t) \in \mathbb{R}_{\leq 1}^+$, $\gamma_{caw}(t) \in \mathbb{R}^+$, and $\Gamma_{cawj}(t) \in \mathbb{R}$ denote the elements of the vectors $\boldsymbol{\beta}_{ca}(t)$, $\boldsymbol{\gamma}_{ca}(t)$, and the matrix $\mathbf{\Gamma}_{ca}(t)$, respectively, where $w, j \in \mathcal{W}_{ca}$.

2.2 Capacities and substitutions

Let $\mathbf{m}(t) \in \mathbb{N}^n$ denote a vector function, where element $m_a(t) \in \mathbb{N}$ defines the inventory capacity of item type a at time t . Further, let the set $\mathcal{B}$ denote the items that are currently out of stock. As a result, $\mathcal{B} \subseteq \mathcal{A}$. If a preferred item is out of stock, then customers of type c will attempt to rent a substitute item $b \in \mathcal{A} \setminus \mathcal{B}$ with probability $r_{cb}(\mathcal{B}) \in \mathbb{R}_{\leq 1}^+$. This paper lets the substitution probability be a function of the set $\mathcal{B}$ to allow customers to prioritize the substitutes. For instance, consider a case of more than two items. In this case, some customers

might want to choose a second alternative if their preferred substitute is out of stock. Note that customers will leave the system if none of the substitutes are in stock. This paper also accounts for the possibility that the rental time distribution when renting the substitute does not equal the distribution of the preferred item. Section 3 delves into the specific details of this behavior.

3 Modeling approach

This section presents a CTMC model of the system described in Section 2.

Let the variable $k_{caw} \in \mathbb{N}_0$ denote the *currently rented* number of items of type $a \in \mathcal{A}$, rented by customers of type $c \in \mathcal{C}$ in phase $w \in \mathcal{W}_{ca}$ of the associated rental time distribution (cf. Section 2.1). Thus, item a can be rented whenever $\sum_{c \in \mathcal{C}} \sum_{w \in \mathcal{W}_{ca}} k_{caw} < m_a(t)$; otherwise item a is out of stock. Now, consider a time-dependent CTMC, where $\mathcal{S}(t)$ denotes the state space at time $t \in \mathbb{R}$, and the state $s \in \mathcal{S}(t)$ denotes a matrix of k_{caw} with an element for each combination of an item type $a \in \mathcal{A}$, customer type $c \in \mathcal{C}$, and phase $w \in \mathcal{W}_{ca}$. Further, let $q_{ss^*}(t) \in \mathbb{R}$ denote the transition rate of the CTMC from the current state s to a new state $s^* \in \mathcal{S}(t)$ at time t . This paper defines the transition rate as follows

$$q_{ss^*}(t) = \begin{cases} \lambda_c(t)\beta_{caw}(t) & \text{if } c = a \text{ and } k_{caw} + 1 \text{ for } s^*, \text{ and } \sum_{c \in \mathcal{C}} \sum_{w \in \mathcal{W}_{ca}} k_{caw} < m_a(t) \text{ for } s. \\ \lambda_c(t)\beta_{caw}(t)r_{ca}(\mathcal{B}) & \text{if } c \neq a \text{ and } k_{caw} + 1 \text{ for } s^*, \text{ and } \sum_{c \in \mathcal{C}} \sum_{w \in \mathcal{W}_{ca}} k_{caw} < m_a(t) \\ & \text{and } \sum_{c \in \mathcal{C}} \sum_{w \in \mathcal{W}_{cb}} k_{cbw} = m_b(t), \text{ where } c = b, \text{ for } s. \\ k_{caw}\Gamma_{cawj}(t) & \text{if } k_{caw} - 1 \text{ and } k_{cawj} + 1, \text{ where } w \neq j, \text{ for } s^*, \text{ and } k_{caw} > 0 \text{ for } s. \\ k_{caw}\gamma_{caw}(t) & \text{if } k_{caw} - 1 \text{ for } s^*, \text{ and } k_{caw} > 0 \text{ for } s. \end{cases}$$

For the diagonal elements, $q_{ss}(t) = \sum_{s^* \in \mathcal{S} \setminus s} q_{ss^*}(t)$; all other rates are zero.

The transitions where $q_{ss^*}(t) = \lambda_c(t)\beta_{caw}(t)$ and $\lambda_c(t)\beta_{caw}(t)r_{ca}(\mathcal{B})$ correspond to customers renting items from the inventory. The first case corresponds to rentals where the customers prefer the items, and the second corresponds to rentals where the items serve as substitutes. The transitions where $q_{ss^*}(t) = k_{caw}\Gamma_{cawj}(t)$ reflect the local transitions from a phase $w \in \mathcal{W}_{ca}$ to a new phase $j \in \mathcal{W}_{ca}$ in the rental time distribution of an ongoing rental. Lastly, the transitions where $q_{ss^*}(t) = k_{caw}\gamma_{caw}(t)$ reflect the items returning to the inventory.

Figure 2 visualizes an example of an isolated part of the transitions in a model containing $n = 2$ types of customers, $\mathcal{C} = \{c_1, c_2\}$, and items, $\mathcal{A} = \{a_1, a_2\}$. The figure illustrates the customers renting item a_1 , and omits the transitions relating to item a_2 . Customers renting item a_1 have the rental time distributions $(\beta_{c_1a_1}(t), \Gamma_{c_1a_1}(t))$ and $(\beta_{c_2a_1}(t), \Gamma_{c_2a_1}(t))$, respectively, where the former represents customers preferring the item, and the latter represents customers using the item as a substitute. The example assumes that both distributions contain two phases, i.e. $\mathcal{W}_{c_1a_1} = \mathcal{W}_{c_2a_1} = \{1, 2\}$.

3.1 Size of the state space

The size of the state space, $|\mathcal{S}(t)|$, is a function of the number of item types, customer types, and the number of phases in the rental time distributions associated with each item. For the computational implementation of the CTMC model, however, one only has to account for the unique distributions within each item type, as the k_{caw} variables can be combined whenever customers renting the same item have identical rental time distributions.

Let $\mathcal{D}$ denote an indexed set for the distributions $(\beta_{ca}(t), \Gamma_{ca}(t))$ unique to each item. Moreover, let the function d denote a mapping $d : \mathcal{D} \rightarrow \mathcal{C} \times \mathcal{A}$ such that $d(i) = i \rightarrow (\mathcal{C} \times \mathcal{A})_i$.

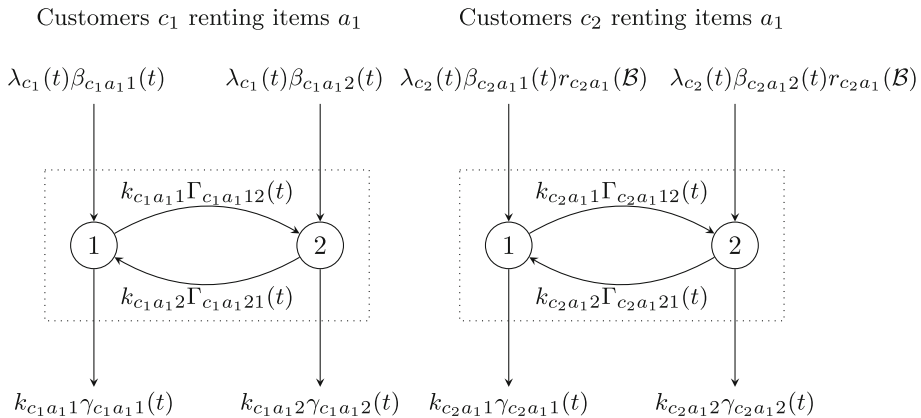


Fig. 2 Example of the transitions in the model for an item type (a_1) rented by two customer types (c_1 and c_2), where the left diagram represents customers preferring the items, and the right, customers using the items as substitutes

In addition, let the function $d(i)_a$ and the set $\mathcal{D}_a \subset \mathcal{D}$ denote the elements that are only related to item a . For example, Figure 2 has $\mathcal{D}_{a_1} = \{1, 2\}$, where $d(1)_{a_1} = c_1 a_1$ (customers c_1 renting items a_1) and $d(2)_{a_1} = c_2 a_1$ (customers c_2 renting items a_1). The computational size of the CTMC model's state space at time t is

$$|\mathcal{S}(t)| = \prod_{a \in \mathcal{A}} \left[\sum_{\kappa_1=0}^{m_a(t)} \sum_{\kappa_2=0}^{m_a(t)-\kappa_1} \cdots \sum_{\kappa_{|\mathcal{D}_a|}=0}^{m_a(t)-\kappa_1-\kappa_2-\cdots-\kappa_{|\mathcal{D}_a|-1}} \prod_{i \in \mathcal{D}_a} \binom{\kappa_i + |\mathcal{W}_{d(i)_a}| - 1}{|\mathcal{W}_{d(i)_a}| - 1} \right]. \quad (1)$$

As a result, a case with $n = 3$ types, where the number of rental time distributions $|\mathcal{D}_a| = 3 \forall a \in \mathcal{A}$, the number of phases $|\mathcal{W}_{d(i)_a}| = 2 \forall i \in \mathcal{D}_a, \forall a \in \mathcal{A}$, and the capacities $m_a(t) = 5 \forall a \in \mathcal{A}, \forall t \in \mathbb{R}$, leads to a state space size of $|\mathcal{S}(t)| \approx 98.6 \cdot 10^6$ states $\forall t \in \mathbb{R}$. The memory required to store the associated state distribution is 788.9 MB, assuming 8 bytes per state.

3.1.1 Special cases

The expression in Equation (1) reduces substantially if one restricts the number of rental time distributions associated with each item. Specifically, if all items include two unique distributions, i.e. $|\mathcal{D}_a| = 2 \forall a \in \mathcal{A}$, then

$$|\mathcal{S}(t)| = \prod_{a \in \mathcal{A}} \binom{m_a(t) + |\mathcal{W}_{d(1)_a}| + |\mathcal{W}_{d(2)_a}|}{m_a(t)}. \quad (2)$$

This special case constitutes all applications where the same rental time distribution governs all customers using item $a \in \mathcal{A}$ as a substitute. Equation (2) reduces if all customers follow the same rental time distribution. In this case, the state space size becomes

$$|\mathcal{S}(t)| = \prod_{a \in \mathcal{A}} \binom{m_a(t) + |\mathcal{W}_a|}{|\mathcal{W}_a|}, \quad (3)$$

where $\mathcal{W}_a$ is the set of phases for the single distribution associated with item $a \in \mathcal{A}$. Equation (3) constitutes all applications where the rental time distribution depends only on the item and not on the customer type.

3.2 Evaluation of the state distribution

This section describes the evaluation approach for the state distribution associated with the aforementioned CTMC.

Let the vector $\pi(t) \in \mathbb{R}^{|\mathcal{S}(t)|}$ denote the state distribution of the CTMC at time $t \in \mathbb{R}$, and let $h \in \mathbb{R}^+$ define the duration of a planning horizon, where $0 \leq t \leq h$. Further, let the planning horizon be divided into several equally sized segments, each with a duration of $\delta \in \mathbb{R}^+$. In addition, let $l = \lceil t/\delta \rceil$ denote the index of a segment, and $\tau = t - \delta(l-1)$ the time spent within the segment associated with l . Further, assume the duration of each segment, δ , is sufficiently small that the difference $|q_{ss^*}(t) - q_{ss^*}(t + \delta)|$ is negligible with respect to the evaluation of $\pi(t)$.

Now consider a transient *homogeneous* CTMC, where $\mathcal{S}_l$ denotes the state distribution of segment l and the vector $\pi_l(\tau) \in \mathbb{R}^{|\mathcal{S}_l|}$ denotes the state distribution at time τ in segment l . Further, let $\mathbf{Q}_l \in \mathbb{R}^{|\mathcal{S}_l| \times |\mathcal{S}_l|}$ denote a matrix, where the elements $q_{ss^*}^l \in \mathbb{R}$ denote the transition rates of the homogeneous CTMC associated with segment l . For the homogeneous parameters, let $q_{ss^*}^l = q_{ss^*}(\psi)$, where $\psi = \delta l$, and initialize each segment letting $\pi_l(\delta) = \pi_{l+1}(0)$ if the state space $\mathcal{S}_l = \mathcal{S}_{l+1}$. Section 3.2.1 describes this paper's approach for initializing the state distribution when $\mathcal{S}_l \neq \mathcal{S}_{l+1}$.

The transient solution to the state distribution of the homogeneous CTMC is

$$\pi_l(\tau) = \pi_l(0)e^{\mathbf{Q}_l\tau}, \quad (4)$$

where $e^{\mathbf{Q}_l\tau} = \mathbf{I} + \sum_{i=1}^{\infty} (\mathbf{Q}_l^i \tau^i)/i!$. The matrix-exponential term can be computationally intractable, even for small state spaces. For this reason, this paper evaluates Equation (4) using the method of uniformization (Ross, 2003, p. 384; Stewart, 2009, p. 361; Bladt and Nielsen, 2017, p. 51), also denoted the method of randomization (Grassman, 1977). The approach employs a discrete-time Markov chain where the time spent in all states is exponentially distributed with rate $\theta_l \in \mathbb{R}^+$. Impose that $\theta_l \geq \max_{s \in \mathcal{S}_l} \{-q_{ss}^l\}$, and let $\mathbf{P}_l = \mathbf{I} + \theta_l^{-1} \mathbf{Q}_l$ denote the transition matrix of the discrete-time Markov chain. Since the time spent in each state of the process is exponentially distributed, the number of transitions after τ units of time, $X \in \mathbb{N}_0$, has a Poisson distribution with rate $\theta_l \tau$. The probability that i transitions have occurred at time τ is therefore $P(X = i) = (\theta_l \tau)^i e^{-\theta_l \tau} / i!$. As a result,

$$\pi_l(\tau) = \sum_{i=0}^{\infty} \pi_l(0) \mathbf{P}_l^i \frac{(\theta_l \tau)^i e^{-\theta_l \tau}}{i!}. \quad (5)$$

Here, $\mathbf{P}_l^i$ is the transition matrix after i transitions of the discrete process. This paper employs the approach provided by Stewart (2009, p. 361) to avoid a direct computation of $\mathbf{P}_l^i$ and $i!$. Firstly, let $\pi_l(\tau) = e^{-\theta_l \tau} \mathbf{x}$, where the value of $\mathbf{x}$ is a result of the infinite series in Equation (5). To derive $\mathbf{x}$, this paper employs an iterative approach that starts by setting $i \leftarrow 0$ and $\mathbf{x} \leftarrow \mathbf{y} \leftarrow \pi_l(0)$. The vector $\mathbf{y}$ is then updated by setting $i \leftarrow i + 1$ and computing $\mathbf{y} \leftarrow \mathbf{y}(\mathbf{P}_l \theta_l \tau / i)$, after which vector $\mathbf{x}$ is updated with $\mathbf{x} \leftarrow \mathbf{x} + \mathbf{y}$.

To determine an appropriate number of iterations, consider a tolerance, $\epsilon \in \mathbb{R}^+$, for the truncation error of the infinite series in Equation (5). Let $\xi \in \mathbb{N}$ denote the number of iterations required to ensure the error is less than ϵ . The number of iterations ξ must be large enough

for $\sum_{i=0}^{\xi} (\theta_l \tau)^i / i! \geq (1 - \epsilon) e^{\theta_l \tau}$ to apply. Once again, this paper avoids direct computation of $i!$ by using an iterative approach that starts by setting $\xi \leftarrow 0$, $z \leftarrow 1$ and $\omega \leftarrow 1$. Next, update with $z \leftarrow z\theta\tau/(\xi + 1)$, $\omega \leftarrow \omega + z$ and lastly with $\xi \leftarrow \xi + 1$. The value of ξ equals the required number of iterations when the inequality $\omega \geq (1 - \epsilon) e^{\theta_l \tau}$ is true.

This paper tests a tolerance of $\epsilon = 1 \cdot 10^{-9}$ throughout all numerical tests (cf. Sections 5–6). In addition, this paper only evaluates the state distribution at the end of each segment — i.e. when $\tau = \delta$.

The reader might have noticed that $e^{-\theta\delta}$ likely leads to numerical underflow, even for small values of the product $\theta\delta$. However, numerical underflow can be avoided by evaluating Equation (5) for a τ smaller than δ and successively re-employing Equation (5) until $\tau = \delta$. Alternatively, the segment duration, δ , can be reduced until Equation (5) can be evaluated directly.

This paper achieves a computationally tractable approach for computing $\pi_l(\tau) \approx \pi(t)$ by segmenting the planning horizon and employing uniformization. The truncation tolerance ϵ and the segment duration δ determines the accuracy of the approach, and will be validated in Section 5.

3.2.1 Change of state space

Consider a case where the subsequent segment leads to a capacity change. In this case, the state space changes, and as a result, the initial state distribution changes as well. Now let $\mathbf{m}_l \in \mathbb{N}^n$ denote a vector, where the element $m_a^l \in \mathbb{N}$ defines the capacity of items of type $a \in \mathcal{A}$ in segment l . If the capacity of an item a increases, i.e. $m_a^{l+1} > m_a^l$, then any states in S_{l+1} where $\sum_{c \in \mathcal{C}} \sum_{w \in \mathcal{W}_{ca}} k_{caw} > m_a^l$ initializes with a probability of zero. On the other hand, if the capacity of an item a decreases, then any states in S_{l+1} where $\sum_{c \in \mathcal{C}} \sum_{w \in \mathcal{W}_{ca}} k_{caw} < m_a^{l+1}$ initializes with a probability equal to the state probabilities in $\pi_l(\delta)$. For the states where $\sum_{c \in \mathcal{C}} \sum_{w \in \mathcal{W}_{ca}} k_{caw} = m_a^{l+1}$, this paper initializes using the corresponding probabilities from $\pi_l(\delta)$ and add the tail probabilities where $\sum_{c \in \mathcal{C}} \sum_{w \in \mathcal{W}_{ca}} k_{caw} > m_a^{l+1}$, assuming the items were removed independently from their current phase.

4 Optimization approach

Rental companies might be confronted with the challenge of estimating the correct capacity by accounting for various aspects of their business. For instance, rental companies can acquire too many or too few items leading to unnecessarily high holding costs or insufficient customer service. Thus, this paper employs the CTMC (cf. Section 3) in a heuristic optimization algorithm and optimizes the capacities of items in the inventory. The paper focuses on items with varying characteristics, yet similar functional properties and, therefore, approximately equal holding cost and price. The items could constitute power drills of varying size or laptops of varying brand. The paper bases the definition of the objective function and constraints on work conducted in collaboration with the rental company Nethire (cf. Section 6).

Let the vector function $\mathbf{f}_a^l(\mathbf{m}_l) \in \mathbb{R}_{<1}^{m_a^l+1}$ denote the marginal occupancy distribution of the CTMC associated with item $a \in \mathcal{A}$ in segment $l \in \{1, 2, \dots, l_{\max}\}$. Recall that $m_a^l \in \mathbb{N}$ denotes an element of vector $\mathbf{m}_l$, defining the inventory capacity of item a in segment l . The parameter $l_{\max} = h/\delta \in \mathbb{N}$ denotes the total number of segments, and function $f_a^{B,l}(\mathbf{m}_l) \in \mathbb{R}_{<1}^+$ the probability that the items of type a are out of stock. The stockout probabilities are a function of the entire vector $\mathbf{m}_l$ since the items can be substituted, and,

therefore, the capacity of one item affects the occupancies of all other items in the system. The proposed optimization algorithm minimizes the sum of inventory capacities, but ensures that the stockout probabilities, $f_a^{B,l}(\mathbf{m}_l)$, do not violate an upper limit, $\rho \in \mathbb{R}_{<1}^+$. Note that $1 - \rho$ reflects the company's service level, i.e. the fraction of customers renting a preferred item. Thus, managers can adjust the value of $1 - \rho$ in line with the company's strategy. Formally, the optimization problem becomes

$$\text{Minimize } \sum_{l=1}^{l_{\max}} \sum_{a \in \mathcal{A}} m_a^l \quad (6)$$

$$\text{Subject to } f_a^{B,l}(\mathbf{m}_l) < \rho \quad \forall a \in \mathcal{A}, \forall l \in \{1, 2, \dots, l_{\max}\} \quad (7)$$

$$m_a^l \in \mathbb{N} \quad (8)$$

where Equation (6) is the objective function, Constraint (7) ensures the stockout probabilities do not violate the upper limit, and Constraint (8) defines the decision variables.

Constraint (7) governs the bound when all parameters of the underlying CTMC model are known. In practice, however, the parameters will be estimated from records of past sales, imposing an uncertainty in the evaluation of $f_a^{B,l}(\mathbf{m}_l)$. If this is the case, Constraint (7) should be validated by estimating the probability $\mathbb{P}(f_a^{B,l}(\mathbf{m}_l) < \rho)$ either during or after the optimization of Equation and Constraints (6)–(8).

The section now continues with a description of the heuristic optimization algorithm.

Let $u_{al}^{\max} \in \mathbb{N}$ and $u_{al}^{\min} \in \mathbb{N}$ denote upper and lower bounds for the capacity m_a^l of item a in segment l . The optimization algorithm consists of an overall procedure that optimizes each segment in ascending order and a nested algorithm that defines the capacity bounds $u_{al}^{\max}$ and $u_{al}^{\min}$.

Algorithm 1 outlines the overall procedure, where Lines 1–2 initialize the fundamental variables, and Lines 3–8 optimize the capacities. Specifically, Line 1 declares the matrix containing the results and initializes the state distribution $\pi_1(0)$ using the currently occupied items. Next, Line 4 derives $u_{al}^{\max}$ and $u_{al}^{\min}$ using the pseudo-code function *getBounds*(π). In Line 5, the pseudo-code function *enumerate*(π, u) optimizes the capacities m_a^l by enumerating the solution space defined by the limits $u_{al}^{\max}$ and $u_{al}^{\min}$. The enumeration procedure evaluates a total of $\prod_{a \in \mathcal{A}} (u_{al}^{\max} - u_{al}^{\min} + 1)$ solutions and returns the capacities that minimize $\sum_{a \in \mathcal{A}} m_a^l$ and respect Constraint (7). Lastly, Lines 6–7 update the state distribution using the pseudo-code function *evaluate*($m[l]$) and increment the segment by one.

Algorithm 2 contains a description of the algorithm that defines the capacity bounds $u_{al}^{\max}$ and $u_{al}^{\min}$. Lines 5–21 alternately decreases the upper bound $u_{al}^{\max}$ and increases the lower bound $u_{al}^{\min}$ until either $u_{al}^{\max} = u_{al}^{\min}$ $a \in \mathcal{A}$ or the bounds no longer improve. Algorithm 2 initializes with $u_{al}^{\max} = 0$ $\forall a \in \mathcal{A}$ and $u_{al}^{\min} = 1$ $\forall a \in \mathcal{A}$ in Line 4. The algorithm finds the upper bound by evaluating each item in sequence (cf. Lines 7–15) and fixing the capacity of the remaining $|\mathcal{A} \setminus a|$ items at the current value of $u_{al}^{\min}$. Correspondingly, the capacity of the item in focus initializes with $u_{al}^{\min}$, but increments by one until $f_a^{B,l}(\mathbf{m}_l) < \rho$, where the pseudo-code function *stockoutProb*(m, a) evaluates the stockout probability $f_a^{B,l}(\mathbf{m}_l)$. Conversely, the algorithm finds the lower bound by fixing the $|\mathcal{A} \setminus a|$ remaining items at the current value of $u_{al}^{\max}$. Lines 16–20 ensure the algorithm alternately improves the two bounds.

The difference $u_{al}^{\max} - u_{al}^{\min}$ is a result of the substitute rentals in the system. Consider an arbitrary item a . The system attains a high rate of substitute rentals when the items in the set $\mathcal{A} \setminus a$ have a capacity equal to the lower bound. Correspondingly, the system attains a low rate when the capacities equal the upper bound. Thus, by improving one bound, the algorithm changes the basis for the value of the other. However, note that improving one bound does

not guarantee a better solution for the other. As a result, the algorithm terminates when either $u_{al}^{max} = u_{al}^{min} \forall a \in \mathcal{A}$, or an improvement could not be obtained.

In the best case, Algorithms 1–2 lead to $u_{al}^{max} = u_{al}^{min} \forall a \in \mathcal{A}$ resulting in a time complexity of $O(n)$. In the worst case, $u_{al}^{max} \neq u_{al}^{min} \forall a \in \mathcal{A}$ resulting in a time complexity of at least $O(2^n)$. Sections 5–6 assess the runtime and solutions derived by the algorithm.

Algorithm 1 Outline of the heuristic optimization algorithm.

```

1:  $m \leftarrow \text{matrix}(l_{max}, n)$ ,  $\pi \leftarrow \text{initStateDist}()$  ▷ Initialize variables.
2:  $l \leftarrow 1$ 
3: while  $l \leq l_{max}$  do
4:    $u \leftarrow \text{getBounds}(\pi)$  ▷ Derive  $u_{al}^{max}$  and  $u_{al}^{min}$  using Algorithm 2.
5:    $m[l] \leftarrow \text{enumerate}(\pi, u)$  ▷ Optimize by enumerating the solution space.
6:    $\pi \leftarrow \text{evaluate}(m[l])$  ▷ Evaluate  $\pi_l(\delta)$  (cf. Section 3.2) using the optimized capacities.
7:    $l \leftarrow l + 1$ 
8: end while
   return  $m$ 

```

Algorithm 2 Algorithm for deriving the capacity bounds u_{al}^{max} and u_{al}^{min} .

```

1: procedure GETBOUNDS
2:    $u \leftarrow \text{matrix}(2, n)$ ,  $m \leftarrow \text{vector}(n)$  ▷ Initialize variables.
3:    $i \leftarrow \text{max}$ ,  $j \leftarrow \text{min}$ 
4:    $u[i] \leftarrow (0, 0, \dots, 0)$ ,  $u[j] \leftarrow (1, 1, \dots, 1)$ 
5:   repeat ▷ Repeat until bounds are equal or did not improve.
6:      $\text{changed} \leftarrow \text{false}$ 
7:     for  $a \in \mathcal{A}$  do
8:        $m \leftarrow u[j]$ ,  $m[a] \leftarrow u[\text{min}][a]$ 
9:       while  $\text{stockoutProb}(m, a) > \rho$  do ▷ Increase capacity until  $\rho$  is no longer violated.
10:         $m[a] \leftarrow m[a] + 1$ 
11:       end while
12:       if  $m[a] \neq u[i][a]$  then ▷ Save the new bound.
13:         $u[i][a] \leftarrow m[a]$ ,  $\text{changed} \leftarrow \text{true}$ 
14:       end if
15:     end for
16:     if  $i = \text{max}$  then ▷ Switch from improving upper bound to lower bound and vice versa.
17:        $i \leftarrow \text{min}$ ,  $j \leftarrow \text{max}$ 
18:     else
19:        $i \leftarrow \text{max}$ ,  $j \leftarrow \text{min}$ 
20:     end if
21:   until  $u[\text{max}] = u[\text{min}]$  or  $\text{changed} = \text{false}$ 
   return  $u$ 
22: end procedure

```

5 Numerical study

This section validates the approaches from Sections 3–4 with the purpose of testing the numerical accuracy of the CTMC approach (cf. Section 3) and the performance of the heuristic optimization algorithm (cf. Section 4).

All experiments implemented the CTMC and heuristic optimization algorithm using the Java programming language (Oracle Corporation, 2025). The implementations are available

on Github¹. The following Section 5.1 compares the CTMC model to various discrete event simulations using the *R* modeling language (R Core Team, 2025).

A computer containing 32 GB of RAM and an AMD EPYC 7502P 32-Core processor with a clock speed of 2.5 GHz conducted all of the following experiments.

5.1 Validation of the CTMC

In this section, the experiments validated the CTMC by comparing the model to discrete event simulations of similar systems. Specifically, the experiments compared the stockout probabilities and expected occupancies of 16 rental systems containing various rental time distributions, customer arrival rates, and substitution patterns.

Let $\tilde{f}_a^l(\mathbf{m}_l) \in \mathbb{R}_{<1}^{m_l^l+1}$ and $\tilde{f}_a^{B,l}(\mathbf{m}_l) \in \mathbb{R}_{<1}^+$ denote the simulated occupancy distribution and stockout probability, respectively. Further, assume the simulation evaluates the system $\eta \in \mathbb{N}$ times and stores the system's state at the end of each segment for a total of $l_{\max}$ segments. The CTMC's stockout probability $f_a^{B,l}(\mathbf{m}_l)$ is statistically equal to the simulated probability, with a significance level of $\alpha \in \mathbb{R}_{<1}^+$, if

$$z_{\alpha/2}\sqrt{l_{\max}^{-1}} < l_{\max}^{-1} \sum_{l=1}^{l_{\max}} \frac{\tilde{f}_a^{B,l}(\mathbf{m}_l) - f_a^{B,l}(\mathbf{m}_l)}{\sqrt{f_a^{B,l}(\mathbf{m}_l)(1 - f_a^{B,l}(\mathbf{m}_l))\eta^{-1}}} < z_{1-\alpha/2}\sqrt{l_{\max}^{-1}}, \quad (9)$$

where $z_{\alpha/2}, z_{1-\alpha/2} \in \mathbb{R}$ denote the standard scores from the normal distribution. In Equation (9), the denominator $\sqrt{f_a^{B,l}(\mathbf{m}_l)(1 - f_a^{B,l}(\mathbf{m}_l))\eta^{-1}}$ evaluates the standard deviation of $f_a^{B,l}(\mathbf{m}_l)$ for η simulations. Thus, for a sufficiently large η , the middle expression follows a normal distribution with a standard deviation of $\sqrt{l_{\max}^{-1}}$. Equation (9) can be adjusted to validate the expected occupancy by replacing $\tilde{f}_a^{B,l}(\mathbf{m}_l) - f_a^{B,l}(\mathbf{m}_l)$ with the difference between the CTMC and the simulated expected occupancy and the product $f_a^{B,l}(\mathbf{m}_l)(1 - f_a^{B,l}(\mathbf{m}_l))$ with the variance of $f_a^l(\mathbf{m}_l)$. This section used both approaches to validate the CTMC.

In addition to Equation (9), this section validated the CTMC by comparing the stockout probability and expected occupancy in each of the $l_{\max}$ segments for a single case. All tests employed a significance level of $\alpha = 0.05$.

The following Sections 5.1.1–5.1.2 describe the input parameters and results of the experiments.

5.1.1 Input parameters

The experiments tested systems containing $n = 2$ and 3 item and customer types with a planning horizon $0 \leq t \leq h = 99$ and segment duration $\delta = 1$, resulting in a total of $h/\delta = l_{\max} = 99$ segments.

In each system, the experiments employed two types of rental time distributions: The Erlang distribution containing 2 phases (also denoted the *Erlang-2*), and the exponential distribution. The experiments kept the parameters of both distributions constant for all segments $l \in \{1, 2, \dots, l_{\max}\}$, item types $a \in \mathcal{A}$, and customer types $c \in \mathcal{C}$, with a mean rental time of 1. The Erlang-2 distribution contained the parameters $\mathcal{W}_{ca} = \mathcal{W} = \{1, 2\}$, the initial distribution $\beta_{ca}(t) = \beta = (1, 0)$ and the PH generator $\Gamma_{ca}(t) = \Gamma$, where the elements

¹ The implementations in Java: github.com/areenberg/TranReloc

$\Gamma_{11} = -2$, $\Gamma_{12} = 2$, $\Gamma_{21} = 0$, and $\Gamma_{22} = -2$. The exponential distribution contained $\mathcal{W} = \{1\}$, $\beta_1 = 1$, and $\Gamma_{1,1} = -1$. Similarly, the experiments employed a constant capacity of $m_a^l = m = 5$ for all segments $l \in \{1, 2, \dots, l_{\max}\}$ and item types $a \in \mathcal{A}$, and an initial occupancy of zero rentals. Further, the experiments contained systems with uniform and zero substitution probability, assuming that each customer only had one substitute. As a result, a customer would be lost from the system whenever the preferred and substitute items were out of stock.

The arrival rates $\lambda_c(t)$ followed a second-degree polynomial with roots in 0 and 100 for all customer types. The experiments contained four maxima ranging from 1 to 4 and assumed that all customer types followed the same arrival pattern — i.e. $\lambda_c(t) = \lambda(t)$. Specifically, for the maximum arrival rate of 1, the experiments used the function $\lambda(t) = -t^2/2500 + t/25$, for a maximum of 2, $\lambda(t) = -t^2/1250 + 2t/25$, for a maximum of 3, $\lambda(t) = -3t^2/2500 + 3t/25$, and lastly for a maximum of 4, $\lambda(t) = -t^2/625 + 4t/25$.

Table 1 provides an overview of the entire setup. Note that a single item reflects the entire system's occupancy, since the constant capacity and customer behavior entails independency between the occupancy distributions and item types.

5.1.2 Results

Figures 3–4 visualizes in details the validation of the system with $n = 2$, the Erlang-2 rental time distribution, and a maximum arrival rate of 1 (cf. Table 1, Test 1). Specifically, Figure 3 validates the stockout probability and Figure 4 the expected occupancy, where the dashed lines reflect the 95% confidence intervals, a solid line reflects the CTMC's estimates, and the dots reflect the simulated estimates. In both cases, the expected number of violations ranges between 1 and 10. Both figures contain 4 violations, which is well within the limits. Table 1 presents the statistical validation of the stockout probability and expected occupancy across all 16 systems. Here, Columns 2–5 contain the experiment setup, and Columns 6–7 the difference between the CTMC and simulation in numbers of standard deviations. Column 8 shows the constant 95% confidence intervals ranging between $-1.97 \cdot 10^{-1}$ and $1.97 \cdot 10^{-1}$. The expected number of violations ranges between 0 and 3 for each measure. Neither the stockout probability nor the expected occupancy violated the limits in any of the 16 cases, indicating that the CTMC does not significantly differ from the simulation.

5.2 Validation of the optimization algorithm

This section validated the optimization algorithm (cf. Algorithms 1–2) by comparing the algorithm to, firstly, complete enumerations of the solution space in six small rental systems, and secondly, a heuristic search procedure in three large rental systems. For the former, the experiments used a lower bound of $u_{al}^{\min} = 1$ and an upper bound $u_{al}^{\max}$ corresponding to the minimal capacity that does not violate an upper stockout probability of ρ for all $a \in \mathcal{A}$ and $l \in \{1, 2, \dots, l_{\max}\}$. The procedure for finding $u_{al}^{\max}$ corresponds exactly to the first iteration of the loop between Lines 5–21 in Algorithm 2. The experiments allowed the enumeration procedure to run without a time limit.

For the three large rental systems, the experiments used a Simulated Annealing (SA) algorithm (Aarts et al., 2014) to search the solution space. Instead of using a prespecified solution space, the experiments initialized the SA from the solution derived by the optimization algorithm and conducted the search with a prespecified time limit. The experiments restarted the search from the initial solution at equal time intervals, where each restart included resetting

Table 1 Result of comparing the CTMC to simulations of 16 rental systems. Columns 2-5 show the setup of each experiment, and Columns 6-7 show the simulated average number of standard deviations from the CTMC's stockout probability and expected occupancy, respectively

#	Nr. items (<i>n</i>)	Max. arr. rate	Rent. dist.	Substitute	Diff. stockout prob.	Diff. occupancy	95% conf. int.
1	2	1	Erlang-2	Yes	$-5.42 \cdot 10^{-2}$	$-1.19 \cdot 10^{-2}$	$(-1.97 \cdot 10^{-1}; 1.97 \cdot 10^{-1})$
2	2	2	Erlang-2	Yes	$3.70 \cdot 10^{-2}$	$-2.96 \cdot 10^{-2}$	$(-1.97 \cdot 10^{-1}; 1.97 \cdot 10^{-1})$
3	2	3	Erlang-2	Yes	$1.63 \cdot 10^{-2}$	$-4.00 \cdot 10^{-2}$	$(-1.97 \cdot 10^{-1}; 1.97 \cdot 10^{-1})$
4	2	4	Erlang-2	Yes	$-1.73 \cdot 10^{-4}$	$7.75 \cdot 10^{-2}$	$(-1.97 \cdot 10^{-1}; 1.97 \cdot 10^{-1})$
5	3	1	Exponential	Yes	$2.98 \cdot 10^{-3}$	$-4.35 \cdot 10^{-2}$	$(-1.97 \cdot 10^{-1}; 1.97 \cdot 10^{-1})$
6	3	2	Exponential	Yes	$-1.07 \cdot 10^{-1}$	$-1.58 \cdot 10^{-1}$	$(-1.97 \cdot 10^{-1}; 1.97 \cdot 10^{-1})$
7	3	3	Exponential	Yes	$-1.01 \cdot 10^{-1}$	$4.87 \cdot 10^{-2}$	$(-1.97 \cdot 10^{-1}; 1.97 \cdot 10^{-1})$
8	3	4	Exponential	Yes	$3.69 \cdot 10^{-2}$	$1.05 \cdot 10^{-1}$	$(-1.97 \cdot 10^{-1}; 1.97 \cdot 10^{-1})$
9	2	1	Erlang-2	No	$5.73 \cdot 10^{-2}$	$-6.36 \cdot 10^{-2}$	$(-1.97 \cdot 10^{-1}; 1.97 \cdot 10^{-1})$
10	2	2	Erlang-2	No	$8.85 \cdot 10^{-2}$	$-4.11 \cdot 10^{-2}$	$(-1.97 \cdot 10^{-1}; 1.97 \cdot 10^{-1})$
11	2	3	Erlang-2	No	$-1.04 \cdot 10^{-1}$	$1.55 \cdot 10^{-2}$	$(-1.97 \cdot 10^{-1}; 1.97 \cdot 10^{-1})$
12	2	4	Erlang-2	No	$7.74 \cdot 10^{-2}$	$1.01 \cdot 10^{-2}$	$(-1.97 \cdot 10^{-1}; 1.97 \cdot 10^{-1})$
13	3	1	Exponential	No	$-3.54 \cdot 10^{-2}$	$4.04 \cdot 10^{-2}$	$(-1.97 \cdot 10^{-1}; 1.97 \cdot 10^{-1})$
14	3	2	Exponential	No	$2.25 \cdot 10^{-2}$	$1.22 \cdot 10^{-1}$	$(-1.97 \cdot 10^{-1}; 1.97 \cdot 10^{-1})$
15	3	3	Exponential	No	$-1.05 \cdot 10^{-1}$	$7.95 \cdot 10^{-3}$	$(-1.97 \cdot 10^{-1}; 1.97 \cdot 10^{-1})$
16	3	4	Exponential	No	$-7.29 \cdot 10^{-2}$	$-9.47 \cdot 10^{-2}$	$(-1.97 \cdot 10^{-1}; 1.97 \cdot 10^{-1})$

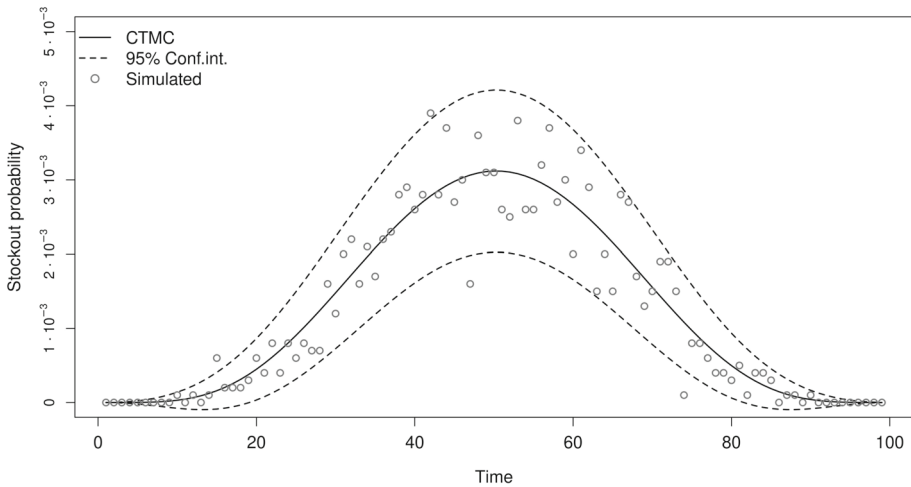


Fig. 3 A comparison of *stockout probabilities* for a single item type from Test 1 (cf. Table 1)

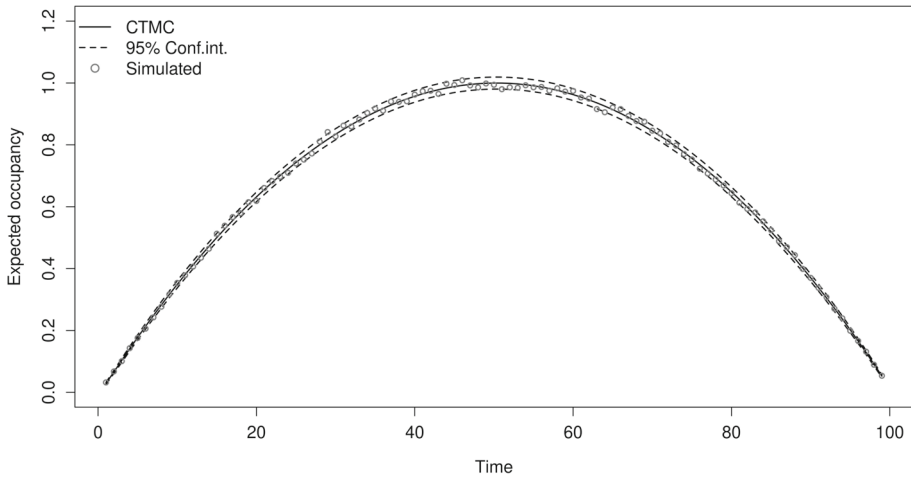


Fig. 4 A comparison of *expected occupancies* for a single item type from Test 1 (cf. Table 1)

the temperature parameter $T \in \mathbb{R}_{\leq 1}^+$. The complement fraction of the remaining time until the next restart defined the temperature. Furthermore, the expression $\exp((\nu' - \nu)/T)$ defined the acceptance probability, where $\nu \in \mathbb{N}$ denotes the objective value (cf. Equation 6) of a candidate solution and $\nu' \in \mathbb{N}$ the objective value of the local best solution.

The experiments validated the performance of the optimization algorithm by taking the sum over the absolute differences $\sum_{l=1}^{l_{\max}} \sum_{a \in \mathcal{A}} |m_a^l - m_a^{*,l}|$, where $m_a^{*,l} \in \mathbb{N}$ denotes the capacity derived by either complete enumeration or SA.

5.2.1 Input parameters

The experiments contained nine rental systems of $n = 2$ and 3 item and customer types. The systems where $n = 2$ employed $l_{\max} = 2$ segments, and the systems where $n = 3$ employed

$l_{max} = 3$ and 20 segments, respectively. All tests used an upper stockout probability of $\rho = 0.01$.

Similar to the previous experiments, items could be substituted with uniform substitution probabilities and customers had only a single substitute. Additionally, the experiments tested the Erlang-2 and exponential rental time distributions, and employed the distributions with a constant mean rental time of 1 in eight systems and means of 1, 1/2, and 1/5 in the ninth system. Furthermore, all customers followed an exponential rental time distribution in the ninth system, independent of the rented item. Thus, for a customer of type c_1 , and items of type a_1, a_2 and a_3 , the distributions $(\beta_{c_1a_1}(t), \Gamma_{c_1a_1}(t)) = (\beta_{c_1a_1}, \Gamma_{c_1a_1}) = (\beta_{c_1a_2}, \Gamma_{c_1a_2}) = (\beta_{c_1a_3}, \Gamma_{c_1a_3})$. The PH generators of the distributions with fractional means were $\Gamma_{ca11} = -2$ for a mean of 1/2, and $\Gamma_{ca11} = -5$ for a mean of 1/5. Section 5.1.1 presents the parameters of the distributions with a mean of 1. Note that phases and initial distributions were independent of the distribution means. Once again, systems where $n = 2$ had all Erlang-2 distributed rental times, and systems where $n = 3$ had all exponentially distributed rental times.

The experiments contained various arrival rate functions. For the three systems where $l_{max} = 2$ segments, all customer types followed the same arrival rate pattern. The experiments firstly tested segments starting and ending with rates of 1/2 and 1. Subsequently, the experiments tested segments containing a constant arrival rate of 1. The systems where $l_{max} = 3$ segments, tested concave, convex, and constant patterns using arrival rates of 1/20 and 1/10, respectively. Once again, all customer types arrived according to the same pattern. For the systems where $l_{max} = 20$, the customer types followed either a concave or convex arrival rate function. The concave case used the function $\lambda_c(t) = -4t^2/441 + 4t/21$, and the convex cases used the function $\lambda_c(t) = |-4t^2/441 + 4t/21 - 1|$.

Finally, the SA search procedure used a time limit of 3 hours and restarted the search once every hour.

Columns 2-6 in Table 2 present an overview of the fundamental input parameters.

5.2.2 Results

Columns 7-11 in Table 2 present the results of the optimization tests. Here, Columns 7-8 contain the runtimes of the optimization algorithm, where Column 7 contains the average runtimes of each segment (corresponding to a single evaluation of Lines 3-8 in Algorithm 1), and Column 8 the total runtimes of the algorithm. The load reduction from Tests 1-3 to Tests 4-6 reduced the average runtimes, whereas the average runtimes increased substantially from Tests 1-3 to Tests 7-9 since customers were given more time to occupy the system. As a result, Tests 7-9 required larger capacities, and thus more iterations, to avoid violating the upper limit ρ .

Columns 9-10 contain the optimized objective values for the algorithm (cf. Column 9), the complete enumerations (cf. Tests 1-6, Column 10), and the SA searches (cf. Tests 7-9, Column 10). Column 11 presents the total absolute differences between the optimized capacities over each segment, showing that the obtained solutions were equal in all tests. Thus, Table 2 indicates that the optimization algorithm derives near-optimal solutions.

6 A company case

This section applies the CTMC to rental data from a company case. Specifically, the section concerns a case of one-way substitution for products from the construction industry.

Table 2 Results of validating the optimization algorithm in nine rental systems. Columns 2–6 contain the experiment setup, where the abbreviations CC and CV denote the concave and convex arrival patterns. Columns 7–11 contain the resulting runtimes and comparisons to complete enumerations (Tests 1–6) and the SA searches (Tests 7–9)

#	Nr. items (n)	Nr. seg. (l_{max})	Arr. rates	Rent. dist.	Rent. means	Avg. rt. (s)	Tot. rt. (s)	Alg. obj. val.	Other obj. val.	Seg. diff.
1	2	2	1/2,1	Erlang-2	1,1	7.43	14.85	14	14	0
2	2	2	1,1/2	Erlang-2	1,1	11.98	23.96	16	16	0
3	2	2	1,1	Erlang-2	1,1	31.22	62.45	18	18	0
4	3	3	1/20,1/10,1/20	Exponential	1,1,1	2.11	6.33	18	18	0
5	3	3	1/10,1/20,1/10	Exponential	1,1,1	2.03	6.10	18	18	0
6	3	3	1/20,1/20,1/20	Exponential	1,1,1	2.04	6.11	18	18	0
7	3	20	CC,CC,CV	Exponential	1,1,1	73.25	1465.09	220	220	0
8	3	20	CC,CV,CV	Exponential	1,1,1	129.54	2590.81	195	195	0
9	3	20	CC,CV,CV	Exponential	1,1/2,1/5	132.87	2657.34	165	165	0

Section 6.1 describes the details of the case, and Section 6.2 validates the adequacy of the CTMC by comparing the marginal occupancy distributions to the observed occupancy over a two-year period. Subsequently, Section 6.3 demonstrates the potential of employing the optimization algorithm by minimizing the capacities in the same period.

6.1 Description of data

The company *Nethire*, a rental service for contractors in the Danish construction industry, provided the data. Nethire allows contractors to rent equipment and machines from a network of various vendors. The investigated case focuses on rentals of scissor lifts from a single vendor from the Copenhagen area. A scissor lift is a mobile platform for lifting workers and equipment to high elevations. It consists of a platform supported by interlocking metal arms that can extend and retract to raise or lower the platform. This case considered scissor lifts of 8 and 12 meters. Customers preferring the 12-meter lift could not use a lift of 8 meters as a substitute since the 8-meter lift was not tall enough. Conversely, customers preferring the 8-meter lift could solve their tasks using the 12-meter lift since the lifts could be lowered. As a result, the scissor lifts only substituted in one direction.

The provided dataset covered transactions of two years, from the 1st of January 2018 up to and including the 31st of December 2019. Each transaction contained the rental date, the return date, and the type of scissor lift. In addition, the company provided information about the vendor's historic capacities. In practice, a rental company would employ a forecast of the demand incorporating analyses of seasonal behaviors. Sections 6.2–6.3 considered the historic two-year period as the planning horizon since the case focuses on validating the adequacy of the CTMC and optimization algorithm.

Let $\mathcal{C} = \{c_{8m}, c_{12m}\}$ denote the two customer types, where c_{8m} denotes the customers preferring a lift of 8 meters and c_{12m} the customers preferring the 12-meter lifts. Correspondingly, let $\mathcal{A} = \{a_{8m}, a_{12m}\}$ denote the set of lift types. In the case of stockout, all customers of type c_{8m} could use lift a_{12m} as a substitute. Thus, the substitution probabilities $r_{c_{8m}a_{12m}}(\mathcal{B}) = 1$, when $\mathcal{B} = \{a_{8m}\}$ was out of stock, and $r_{c_{12m}a_{8m}}(\mathcal{B}) = 0$, when $\mathcal{B} = \{a_{12m}\}$ was out of stock. Additionally, customers of type c_{8m} retained their rental time distribution when renting the substitute lift a_{12m} . Since the lifts substituted in one direction, $|\mathcal{D}_{a_{8m}}| = 1$ distribution associated with lift a_{8m} , denoted $(\beta_{c_{8m}a_{8m}}(t), \Gamma_{c_{8m}a_{8m}}(t))$, and $|\mathcal{D}_{a_{12m}}| = 2$ distributions associated with lift a_{12m} , denoted $(\beta_{c_{12m}a_{12m}}(t), \Gamma_{c_{12m}a_{12m}}(t))$ and $(\beta_{c_{8m}a_{12m}}(t), \Gamma_{c_{8m}a_{12m}}(t))$, where $(\beta_{c_{8m}a_{12m}}(t), \Gamma_{c_{8m}a_{12m}}(t)) = (\beta_{c_{8m}a_{8m}}(t), \Gamma_{c_{8m}a_{8m}}(t))$. Thus, the entire system contained a total of two unique distributions. The distributions could be described by general PH distributions containing $|\mathcal{W}_{c_{8m}a_{8m}}| = |\mathcal{W}_{c_{12m}a_{12m}}| = |\mathcal{W}_{c_{8m}a_{12m}}| = 2$ phases. Further, the distributions followed a step function, resulting in homogeneous parameters in periods of six months.

This section estimated the parameters of the PH distributions using the Expectation-Maximization (EM) algorithm from the *EMph* program (Asmussen et al., 1996, 1998). Specifically, the EM algorithm estimated the homogeneous parameters of each six-month period, leading to four estimates of each parameter in the PH generators and initial distributions, respectively. Table 3 contains the results for both lift types. In addition, Table 3 shows the means and standard deviations associated with each set of parameters, showing that lifts experienced long rental times from January to June and short rental times from July to December in both years.

This section based the arrival rate functions, $\lambda_{8m}(t)$ and $\lambda_{12m}(t)$, on Poisson regressions of the observed daily number of rentals. Specifically, the planning horizon divided the 8-meter

lifts into two intervals, defining the first from the 1st of January to the 30th of September 2018 and the second from the 1st of October 2018 to the 31st of December 2019. Each interval used second-order polynomials to model the behavior of the regression response variable. Fitting the parameters in the R modeling language led to the expression $\log(\lambda_{8m}(t)) = 9.14 \cdot 10^{-5}t^2 - 3.06 \cdot 10^{-2}t - 9.49 \cdot 10^{-1}$ for $0 \leq t < 273$ and $\log(\lambda_{8m}(t)) = 1.06 \cdot 10^{-5}(t - 273)^2 - 8.13 \cdot 10^{-3}(t - 273) - 6.95 \cdot 10^{-1}$ for $273 \leq t \leq 730$. Conversely, the response variable of the 12-meter lift followed a second-order polynomial stretching across the entire two-year planning horizon. Fitting the parameters in R led to the expression $\log(\lambda_{12m}(t)) = -8.02 \cdot 10^{-6}t^2 + 8.10 \cdot 10^{-3}t - 3.48$ for $0 \leq t \leq 730$.

Figure 8 visualizes the modeled and observed mean daily number of arrivals, and Figure 9 visualizes the historic capacities. The system initialized with occupancies of zero, and used a segment duration of $\delta = 1$ day, leading to a total of $l_{max} = 730$ segments.

6.2 Validation of the CTMC

The following validates the CTMC by evaluating the marginal occupancy distributions of each lift. Figure 5 compares the 2.5% and 97.5% percentiles of the CTMC's marginal occupancy distributions to the observed occupancy in our data. If the CTMC adequately reflects the observed occupancy and the violations of the percentiles occur independently, one would observe between 25 and 48 violations with a significance level of 5%. However, due to the observations' mutual dependence, the true limits are substantially greater. Executing 10000 simulations of the two-year planning horizon yields an interval of 0 to 61 violations for the 8-meter lifts and 0 to 49 violations for the 12-meter lifts. The observed occupancy (cf. Figure 5) has 1 violation for the 8-meter lifts and 13 violations for the 12-meter lifts, indicating that the CTMC adequately reflects the data.

6.3 Optimization of the inventory

This section demonstrates the potential of the proposed optimization algorithm by minimizing the vendor's inventory. The section aimed to minimize the capacity of the vendor's inventory without violating the upper stockout probability ρ at any time in the planning horizon. Similar to Section 5.2, the target upper stockout probability equaled $\rho = 0.01$. Historically, the company's maximum stockout probability equaled $2.12 \cdot 10^{-2}$ for the 8-meter lifts and $1.57 \cdot 10^{-1}$ for the 12-meter lifts.

Figure 6 visualizes the optimized and actual capacities of the two lifts. The algorithm runtime was 18.70 seconds on average in each segment and 13651.37 seconds in total. The maximum estimated stockout probabilities in the optimized solution were $9.99 \cdot 10^{-3}$ for the 8-meter lifts and $9.97 \cdot 10^{-3}$ for the 12-meter lifts. For the validation of these results, Figure 7 visualizes the uncertainty associated with the stockout probability, estimated by parametric bootstrapping. Figure 7 shows that the true stockout probability is less than 0.1 with a probability of at least 95% for the entire planning horizon.

The optimized solution reduced the estimated stockout probabilities and the objective value from 8642 to 7168 item days, corresponding to a reduction of 17.06%. Individually, the reductions were 29.27% and 4.50% for the 8 and 12-meter lifts, respectively. Thus, if the holding costs per item were constant for the entire two-year period, the savings of employing the solution would have been substantial. The solution even improved customer service by reducing the maximum stockout probabilities.

Table 3 Rental time distributions for customers renting the two lift types. Columns contain the distribution parameters ($\beta_{ca}(t)$, $\Gamma_{ca}(t)$), mean, and standard deviation (in days) for each period of six months

	Scissor lifts of 8 meters (8m)				Scissor lifts of 12 meters (12m)			
	Jan-Jun (2018)		Jul-Dec (2018)		Jan-Jun (2019)		Jul-Dec (2019)	
	Jan-Jun (2018)	Jul-Dec (2018)	Jan-Jun (2018)	Jul-Dec (2018)	Jan-Jun (2019)	Jul-Dec (2019)	Jan-Jun (2019)	Jul-Dec (2019)
$\Gamma_{ca11}(t)$	$-5.93 \cdot 10^{-2}$	-2.86	$-3.98 \cdot 10^{-1}$	-2.34	$-7.70 \cdot 10^{-1}$	$-3.73 \cdot 10^{-1}$	$-2.79 \cdot 10^{-1}$	$-5.93 \cdot 10^{-1}$
$\Gamma_{ca12}(t)$	$3.89 \cdot 10^{-2}$	$7.13 \cdot 10^{-1}$	$6.37 \cdot 10^{-2}$	$6.13 \cdot 10^{-1}$	$1.80 \cdot 10^{-1}$	$4.66 \cdot 10^{-2}$	$4.41 \cdot 10^{-2}$	$1.04 \cdot 10^{-1}$
$\Gamma_{ca21}(t)$	$2.25 \cdot 10^{-1}$	$4.58 \cdot 10^{-1}$	$2.07 \cdot 10^{-2}$	$5.08 \cdot 10^{-1}$	$1.53 \cdot 10^{-2}$	$2.00 \cdot 10^{-2}$	$2.62 \cdot 10^{-2}$	$9.53 \cdot 10^{-2}$
$\Gamma_{ca22}(t)$	$-6.98 \cdot 10^{-1}$	$-4.58 \cdot 10^{-1}$	$-2.73 \cdot 10^{-2}$	$-5.08 \cdot 10^{-1}$	$-1.53 \cdot 10^{-2}$	$-2.96 \cdot 10^{-2}$	$-4.05 \cdot 10^{-2}$	$-1.98 \cdot 10^{-1}$
$\beta_{ca1}(t)$	$1.19 \cdot 10^{-1}$	0.00	$9.07 \cdot 10^{-1}$	0.00	$9.99 \cdot 10^{-1}$	$8.91 \cdot 10^{-1}$	$8.26 \cdot 10^{-1}$	$5.12 \cdot 10^{-1}$
$\beta_{ca2}(t)$	$8.81 \cdot 10^{-1}$	1.00	$9.34 \cdot 10^{-2}$	1.00	$8.61 \cdot 10^{-4}$	$1.09 \cdot 10^{-1}$	$1.74 \cdot 10^{-1}$	$4.88 \cdot 10^{-1}$
Mean	10.39	3.38	12.73	3.25	21.69	10.93	12.12	4.57
Stand. dev.	17.71	3.06	27.60	2.88	55.44	23.51	20.47	5.28

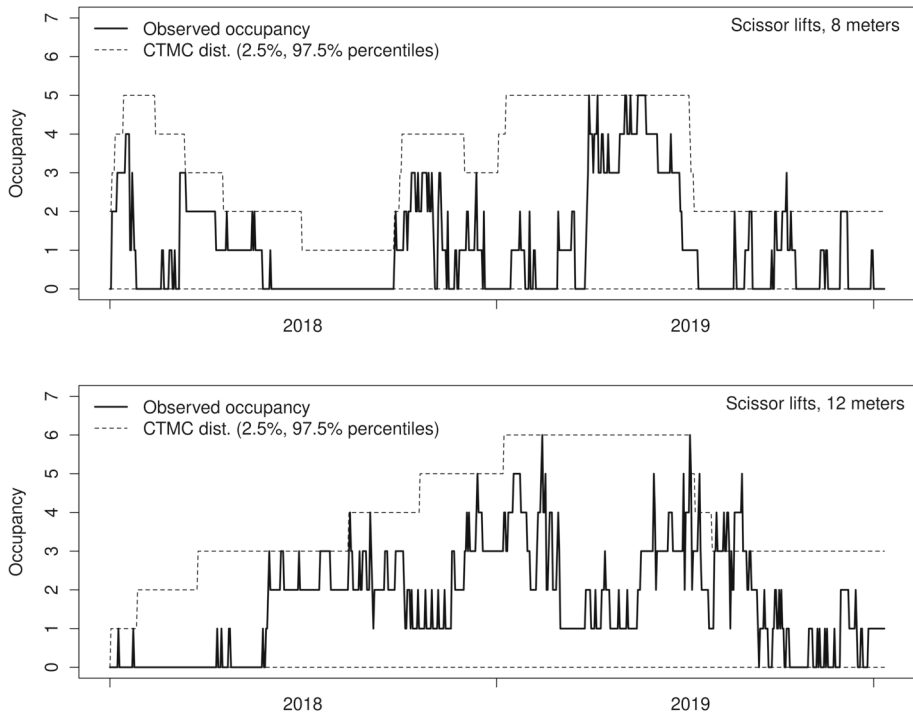


Fig. 5 The observed number of occupied items, including percentiles from the CTMC's marginal occupancy distributions. Displays scissor lifts of 8 meters (top) and 12 meters (bottom)

7 Conclusion

This paper proposes a set of methods for evaluating and optimizing an inventory considering a time-dependent rental system with substitutable items. The paper assumes that customers arrive according to time-dependent Poisson processes and follow phase-type distributed rental times. Based on a CTMC, the paper evaluates the time-dependent behavior by segmenting the planning horizon and employing the uniformization method. This approach results in an occupancy distribution for each item. Moreover, the paper provides a heuristic optimization algorithm that minimizes the total capacities in the planning horizon but ensures the stockout probabilities do not violate an upper limit.

The numerical experiments in Section 5 showed no significant difference between the CTMC and a discrete event simulation of the rental system, indicating that the implementation adequately evaluates the occupancy distributions of the CTMC. Similarly, there was no difference between the solutions obtained by the optimization algorithm and those obtained by complete enumeration or simulated annealing. Thus, the experiments indicated that the algorithm leads to near-optimal solutions. Finally, Section 6 showed that by employing the algorithm, a vendor could reduce the inventory capacities and simultaneously the maximum stockout probabilities, improving customer service. Altogether, Sections 5–6 indicate that the proposed methods can be useful for rental companies aiming to reduce their inventory.

Future research should focus on reducing the computational memory requirements of the CTMC model. This can be achieved by exploiting the numerical advantages of PH distributions, such as conditions where the representation reduces to the exponential distribution

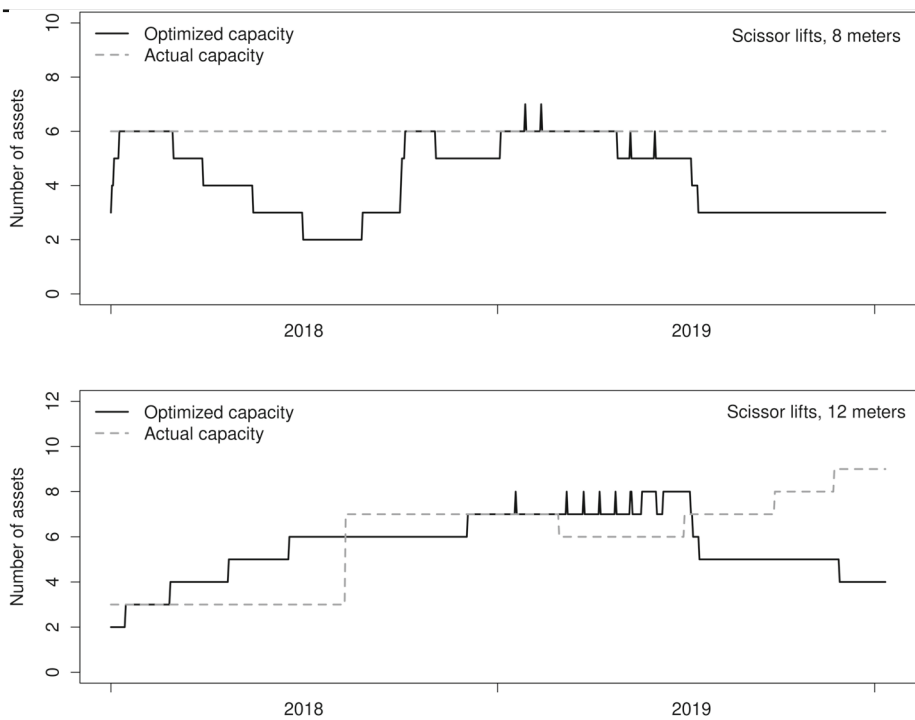


Fig. 6 Optimized ($\rho = 0.01$) and actual capacities from the company case. Displays scissor lifts of 8 meters (top) and 12 meters (bottom)

(Gardner, 2024). In other cases, it might be possible to approximate the density by a sub-class of PH distributions containing fewer parameters. Furthermore, stationary systems, where the arrivals follow Poisson processes, are inherently insensitive to the type of rental time distribution, revealing another opportunity for reducing the model (Boucherie & Dijk, 2011).

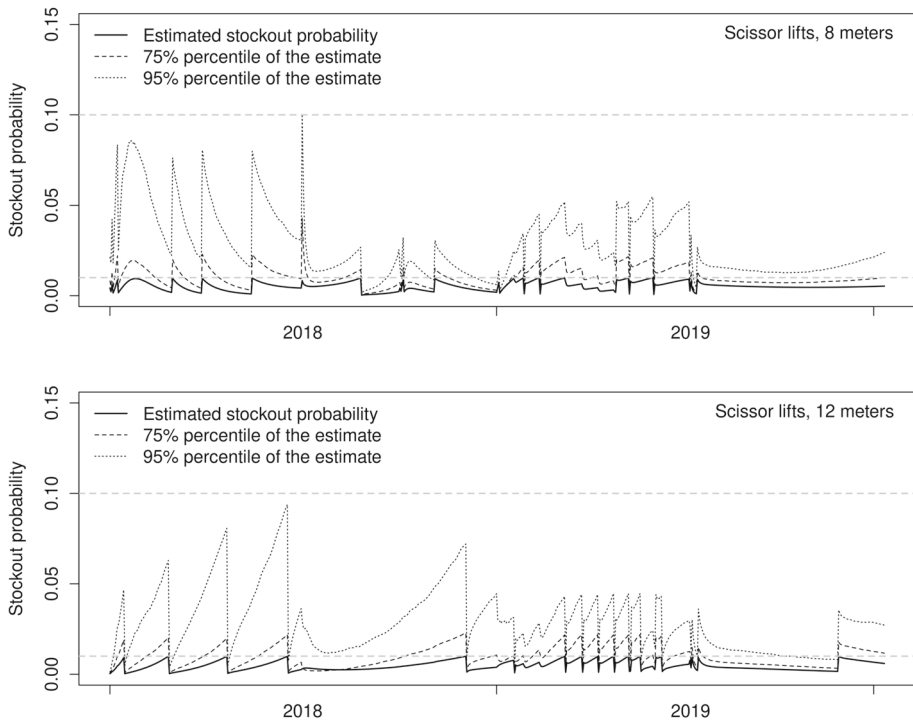


Fig. 7 The estimated stockout probability, including 75% and 95% percentiles for the distribution of the estimate. The grey horizontal lines mark the stockout probabilities of 0.01 and 0.1. Displays scissor lifts of 8 meters (top) and 12 meters (bottom)

Appendix A Extended data description

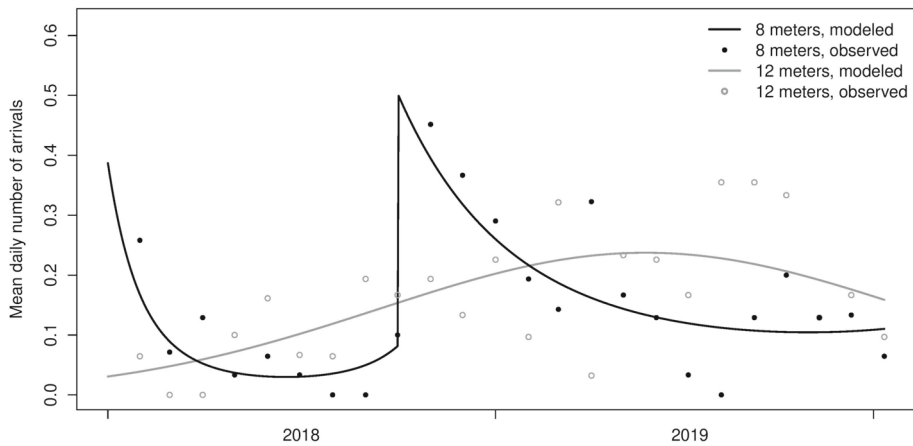


Fig. 8 Modeled and observed mean daily number of arrivals for customers preferring the lifts from the company case. The dots represent the observed mean in each month

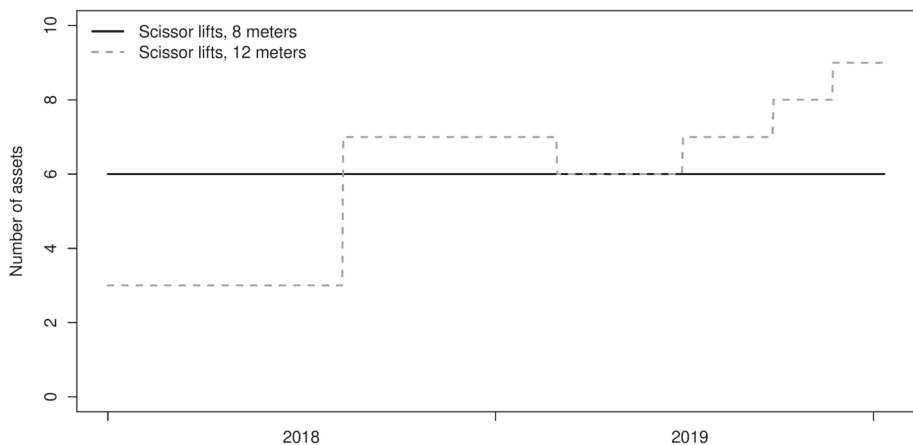


Fig. 9 Historic capacities for the lifts from the company case

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Declarations

Competing Interests Not applicable.

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